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PFM-1 Landmine Detection in Vegetation Using Thermal Imaging with Limited Training Data

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Abstract: Landmine detection, especially of PFM-1 "butterfly" mines, remains a critical challenge in post-conflict environments due to their small size and plastic construction. Modern demining operations increasingly incorporate advanced sensing modalities, including multi-spectral imaging. Long-Wave Infrared (LWIR), in particular, offers potential for detecting PFM-1 mines under certain environmental conditions. However, the limited availability of annotated thermal data restricts the generalization capabilities of deep learning approaches. To address these limitations, we propose a multi-stage, feature-based detection algorithm tailored for PFM-1 mines in LWIR thermal imagery. The method is trained on Track 1 and Track 2 of the MineInsight dataset, using F1 and F2 score-weighted loss functions to prioritize recall and reduce false negatives. Across ten independent runs, the proposed method exhibits comparable performance and complementary robustness relative to YOLOv8, especially under conditions of limited training data or partial occlusion.

Keywords: Humanitarian demining, landmine detection, long-wave infrared, thermal anomaly analysis, field robotics.

1. INTRODUCTION

Landmines remaining from past conflicts pose severe, long-term threats to civilian populations and hinder regional socioeconomic recovery. Manual demining is slow and hazardous, driving the adoption of robotic systems to improve operational safety and efficiency. Funded initiatives [1,2] aim to standardize robotic demining practices. Concurrently, research has increasingly explored equipping these robotic platforms with diverse sensing technologies beyond conventional visual feedback, aiming to enhance landmine detection through multi-spectral imaging capabilities [3–7].

The PFM-1 "butterfly" anti-personnel (AP) landmine presents a significant detection challenge due to its small dimensions and being mainly made of plastic. Among various sensing modalities, Long-Wave Infrared (LWIR) thermal imaging has shown notable potential for PFM-1 identification in specific hours of the day [3]. Other approaches [5] [8] use deep learning techniques, which typically require large, diverse datasets for effective training and generalization. As highlighted by Baur et al. [8], even deep learning models can exhibit significant performance degradation when exposed to unseen data or variations in environmental conditions. This indicates challenges in model generalization, particularly when datasets are limited or subject to domain shifts. This limitation is particularly critical for PFM-1 thermal detection, where few datasets are available. Such data scarcity restricts the application of data-intensive deep learning models and underscores the need for alternative or complementary algorithmic strategies.

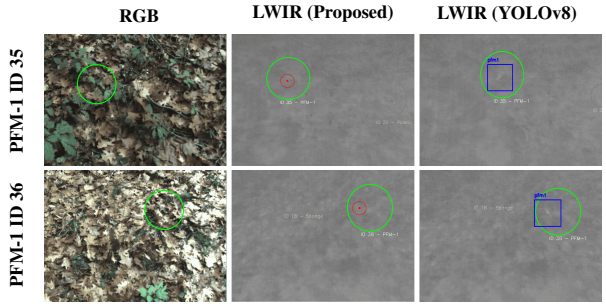


Fig. 1. Qualitative comparison of PFM-1 detection on Track 2 of the MineInsight [9] dataset (samples ID 35/36, varying leaf occlusion). Left: RGB reference with ground truth (green). Middle: LWIR frames from a Teledyne FLIR Boson 640 camera; green = ground truth, red = proposed method. Right: same LWIR frames with YOLOv8 predictions (blue). We show the two best-performing models for qualitative assessment.

To address these challenges in PFM-1 detection using thermal imaging, particularly in scenarios constrained by limited data availability for deep learning applications, this paper proposes and evaluates a multi-stage feature-based algorithm. Designed for PFM-1 landmine detection using LWIR thermal imagery acquired by an Unmanned Ground Vehicle (UGV), this algorithm demonstrates applicability in challenging environments characterized by moderate vegetation. A primary contribution of this work is the demonstration that targeted hyperparameter optimization enhances detection performance, especially under conditions of limited data

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availability that typically constrain deep learning approaches. We also benchmark its performance against the YOLOv8 [10] architecture. Representative detection results from both methods, under varying occlusion conditions, are shown in Figure 1.

2. RELATED WORK

The detection of small plastic AP landmines like the PFM-1 is particularly critical in environments such as moderately or sparsely vegetated rubble fields, which are common in many post-conflict zones.

Nikulin et al. [3] investigated the feasibility of detecting these mines using UAV-based thermal imaging. In their study, they demonstrated that thermal properties of the plastic casing produce detectable anomalies relative to the surrounding ground, especially during periods of significant temperature change, such as post-sunrise. Their field trials, conducted over simulated minefields in environments including low grass and rubble, confirmed that PFM-1 mines could be reliably detected and identified using thermal data collected from a low-altitude UAV, achieving detection sensitivities over 77% during early morning flights. However, performance may be limited in areas with dense vegetation.

Building upon these findings, Baur et al. [8] focused specifically on automating the detection of the PFM-1 mines using deep learning. They collected multispectral and thermal infrared datasets using a UAV over simulated minefields containing PFM-1 mines and their carrying case in rubble, grass, and snow environments. After processing the data in georeferenced orthophotos, they applied a Faster R-CNN [11] object detection model. Training was performed on augmented datasets derived from the orthophotos. Their results showed 99.3% mean Average Precision (mAP) on randomized train/test splits from the same flights. However, mAP dropped sharply to 71.5% when testing the model trained on 2019 data against a similar environment surveyed in 2017 (a fully withheld dataset). This decrease highlighted the challenge of model generalization across different times and potentially subtle environmental changes, even when using advanced models like Faster R-CNN for PFM-1 detection.

The high cost and bulkiness of high-resolution thermal cameras often represent an additional challenge. In [4], Lin and Wang proposed the MineSeeker system to specifically tackle PFM-1 detection using affordable, low-resolution thermal cameras. They first pre-process the low-resolution thermal data through temperature filtering and color mapping, and then enhance the image using a combination of cubic spline and bicubic interpolation. The enhanced image is then fed into a YOLOv8 deep learning model for object detection. Their experiments demonstrated that pre-processing improved the detection performance of PFM-1 (mAP50-95 increased from 55% to 78%) compared to applying YOLOv8 directly to raw low-resolution thermal data.

Vivoli et al. [5] used lightweight YOLOv8 models (nano and small) for real-time detection of surface mines (PFM-1 and PMA-2) using a mobile platform. Starting from their SULAND dataset, collected under various environmental conditions, they achieved high recall (missing less than 1.6% of targets) on in-distribution test data. However, false positive rates exceeded 30%, which they deemed acceptable as detections primarily served to guide a human operator via a remote interface.

Recently, significant progress was made in landmine detection, especially for difficult targets like the PFM-1. While deep learning is now a prominent technique for computer vision, it still faces challenges in generalizing across environments and requires large, diverse training datasets. Our work investigates a UGV-based system using a LWIR thermal camera and robotic arm for PFM-1 detection. In contrast to deep learning-based methods, we propose a feature-driven detection algorithm with targeted hyperparameter tuning, designed for deployment under data-scarce and partially occluded conditions.

3. METHODS

This section presents the two thermal mine detection approaches evaluated in this study: a feature-based method and a YOLOv8-based deep learning model.

3.1 Proposed Method

We propose a thermal camera processing pipeline that combines pixel-level anomaly analysis, blob filtering, and strength-based detection logic. In the following, we briefly summarize each of these stages.

Initial Thermal Signature Extraction

We extract a pseudo-thermal signature from the LWIR data, visualized using an RGB JET color map:

$$I_{\text{sig}}(x,y) = \left[I_R(x,y) - I_B(x,y)\right]_0^{255} \tag{1}$$

where I_R and I_B are the red and blue channel intensities of the image, respectively. The operation $[\cdot]_0^{255}$ denotes clipping into the range $[0, 255]$, removing negative artifacts. We later obtain I_{norm} by normalizing I_{sig} to the range $[0, 255]$. We then compute a temporal persistence mask M_p over the last N_H frames to suppress transient noise by retaining only stable responses:

$$M_p(x,y) = \begin{cases} 1, & \sum[I_{\text{norm}}^{t-i}(x,y) > T_{\text{hist}}] \geq N_H - 1 \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

Here, T_{hist} is the persistence threshold. We assume minimal camera motion over short windows, such that pixel-wise comparisons across frames remain valid without explicit motion compensation. In our setup, the LWIR camera is rigidly mounted to a slowly moving UGV with limited vibration, supporting this assumption empirically.

The filtered image is then given by:

$$I_{\text{filt}}(x,y) = I_{\text{norm}}(x,y)\, M_p(x,y) \tag{3}$$

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Adaptive Binarization and Morphological Operations

An adaptive threshold, T_{adapt}, is obtained as the P_{hot}-th percentile of the non-zero intensities in I_{filt}:

$$T_{\text{adapt}} = \text{percentile}\big(\{v_j\},\, 100 - P_{\text{hot}}\big) \tag{4}$$

where $\{v_j\}$ denotes all positive pixel values in I_{filt}. This percentile-based rule emphasizes high-contrast regions, assuming that candidate anomalies exhibit stronger thermal signatures than the background. The algorithm then binarizes the image:

$$I_{\text{bin}}(x,y) = \begin{cases} 255, & I_{\text{filt}}(x,y) \geq T_{\text{adapt}} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

Thermal camera speckle noise is suppressed by a sequence of morphological erosion and dilation:

$$I_{\text{dilated}} = \big(I_{\text{bin}} \ominus K_{S_{\text{erode}}}\big) \oplus K_{S_{\text{dilate}}} \tag{6}$$

with K_n an $n \times n$ square structuring element. S_{erode} and S_{dilate} control each operation's extent. After morphology, we extract connected components (blobs) from I_{dilated}; blobs with area $A < A_{\min}$ are discarded. For each remaining blob B, the thermal contrast with respect to a surrounding ring of radius r_{ring} is given by:

$$\Delta G = \left| \overline{I}_B - \overline{I}_{\text{ring}} \right| \tag{7}$$

where $\overline{I}_B$ and $\overline{I}_{\text{ring}}$ are the mean $R-B$ values within the blob B and its ring of radius r_{ring}, respectively.

Core Detection Logic

Blobs in the current frame are associated with those in the previous frame by minimizing centroid distance and then updating (or initializing) a strength score S. Three mutually exclusive scenarios arise:

Matched blobs: For every blob in frame $t-1$, locate the closest blob in frame t whose centroid distance is below $D_{\max}$; otherwise, repeat the search within a fallback radius R_{fb}. After association, append the new gradient to the history (length N_g) and compute the mean gradient $\bar{G}$. Here, T_G is the minimum contrast threshold for valid blobs.

$$\bar{G} \geq T_G \quad \Longrightarrow \quad S_{\text{new}} = 1 \tag{8a}$$

Otherwise:

$$S_{\text{new}} = S_{\text{old}} \Big(M_{\text{base}} + M_{\text{bonus}} \min\big(1, \tfrac{\bar{G}}{T_G}\big) \Big) \tag{8b}$$

where M_{base} and M_{bonus} are the base and bonus decay coefficients, respectively. If $S_{\text{new}} < T_{\text{pers}}$, the blob's persistence flag is reset; otherwise, the flag remains set.

Unmatched blobs: If a blob fails to match in the current frame, its strength decays according to:

$$S_{\text{temp}} = S_{\text{old}} \Big(U_{\text{base}} + U_{\text{bonus}} \min\big(1, \tfrac{\bar{G}}{T_G}\big) \Big)\, D_S \tag{9}$$

Here, U_{base} and U_{bonus} are the base and bonus decay coefficients for unmatched blobs, and D_S is an overall decay factor. If $S_{\text{temp}} \geq S_{\min}$ (minimum strength threshold) attempt re-association within R_{fb} using a further-decayed gradient D_{grad}; otherwise drop the blob.

New blobs: When a blob first appears, set its initial strength S_{init} as:

$$S_{\text{init}} = \begin{cases} S_{\text{init}}^{\text{valid}}, & \Delta G \geq T_G, \\ S_{\text{init}}^{\text{invalid}}, & T_G V_{\text{margin}} < \Delta G < T_G, \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where V_{margin} is the margin ratio for an intermediate gradient. If $S_{\text{init}} > 0$ persistence starts "on"; otherwise it remains "off."

Finally, a blob becomes a qualified detection once its current strength meets:

$$S_{\text{current}} \geq S_{\min} \tag{11}$$

The centroid (x, y) of every qualified detection is then stored for subsequent evaluation.

Parameter Tuning

We tuned the parameters of our method with the Optuna [12] optimization framework. The variables exposed to the optimizer, grouped as $\mathbf{P}_{\text{tuned}}$, are listed in Table 1; their search ranges are detailed in Section 3.4.

We set the remaining parameters manually. The morphological kernel sizes S_{erode} and S_{dilate} and the ring radius r_{ring} were adjusted to match the nominal camera–track distance. We fixed the thermal-related parameters T_{hist}, N_H, and N_G according to the ambient conditions and the sensor refresh rate.

3.2 YOLOv8

For comparative evaluation, we employ the YOLOv8l (large) model [10], a CNN-based detector from the YOLO family [13], known for its favorable accuracy–speed trade-off, mature and widely adopted implementation. We adapt input thermal images to the model's three-channel requirement through single-channel replication. We initialize the network with pre-trained YOLOv8l (large) weights, freeze the first ten backbone layers, and fine-tune only the detection head. We train our model using the Ultralytics framework and implement a custom data augmentation strategy—incorporating contrast adjustment, rotation, scaling, and tiling—which showed better efficacy on our thermal images over default YOLOv8 augmentations. We propose two methods based on YOLOv8l (large), namely YOLOv8 P1 and YOLOv8 P2, where P1 and P2 refer to parameter sets 1 and 2, respectively. As their name suggests, those two methods only differ by the hyperparameters used to fine-tune the pre-trained YOLOv8l model. Details on those two parameter sets can be found in Section 3.4.

3.3 Dataset

We conducted our study using the MineInsight dataset [9]. Although it includes dual-view sensor scans from a UGV and its robotic arm, along with multiple camera modalities, we focused exclusively on LWIR data captured by a Teledyne FLIR Boson 640 camera. We analyzed two sequences: Track 1, Sequence 1 (denoted as T1), collected between 1:00 PM and 2:00 PM, with

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7635 LWIR frames and 9 PFM-1 mines, and Track 2, Sequence 1 (denoted as T2), collected between 3:00 PM and 4:00 PM, with 6732 LWIR frames and 6 PFM-1 mines. During collection, the UGV moved at $\approx 0.2\,\mathrm{m/s}$ with its robotic arm fully extended and tilted 45° downward. The FLIR camera's position resulted in an approximate height of $1.7\,\mathrm{m}$ above ground level throughout the sequences, with data streamed at $30\,\mathrm{Hz}$. As the camera is rigidly mounted to the UGV, it shares the same velocity as the platform with respect to the terrain. Based on the platform speed ($\approx 0.2\,\mathrm{m/s}$), camera height ($\approx 1.7\,\mathrm{m}$), and a 95° horizontal field of view, the apparent motion of ground features is ≈ 1.15 pixels per frame at $30\,\mathrm{Hz}$. While the mounting is fixed, minor height variations may occur due to terrain irregularities or small bumps along the track. For detailed information on the dataset and data collection method, we refer to the MineInsight article [9].

3.4 Training

The MineInsight dataset comprises an estimation of the location of the targets that can be used as ground truth for learning algorithms. We train both algorithms first on T1 and tested on T2 (T1→T2) and then on T2 with test on T1 (T2→T1). Training was performed on an AMD Threadripper PRO 7985WX (64 cores).

To tune our method, we constrain the search space of $\mathbf{P}_{\text{tuned}}$, with ranges chosen based on domain knowledge and empirical constraints. Table 1 reports the parameter bounds used for optimization.

Table 1. Definition of the hyperparameter search space for the Optuna-based optimization.

Parameter	Range	Parameter	Range
P_{hot}	$[0.1,\ 0.5]$	M_{base}	$[0.6,\ 0.8]$
A_{min}	$[15,\ 40]$	M_{bonus}	$[0.2,\ 0.5]$
T_G	$[60,\ 80]$	U_{base}	$[0.4,\ 0.7]$
D_{max}	$[30,\ 60]$	U_{bonus}	$[0.1,\ 0.4]$
S_{min}	$[0.4,\ 0.95]$	V_{margin}	$[0.5,\ 0.9]$
D_S	$[0.4,\ 0.9]$	D_{grad}	$[0.0,\ 1.0]$
T_{pers}	$[0.2,\ 1.0]$	R_{fb}	$[10,\ 40]$
$S_{\text{init}}^{\text{valid}}$	$[0.7,\ 1.0]$	$S_{\text{init}}^{\text{invalid}}$	$[0.3,\ 0.8]$

We trained ten independent instances of our algorithm for each optimization objective, initially maximizing the F1 score (denoted as Ours F1). Each instance was trained with $n = 2500$ Optuna trials for both directions: T1→T2 and T2→T1. Given that evaluation is performed on a per-frame basis, a mine must be consistently detected throughout its visible duration. For instance, if a PFM-1 mine is correctly detected as a true positive (TP) in only $n\%$ of the frames in which it appears, it will be counted as a false negative (FN) in the remaining $(100 - n)\%$ frames. This naturally results in a high FN count. To address this and prioritize recall in our final metrics, we also optimized using the F2 score (denoted as Ours F2), which assigns more weight to recall relative to precision. Our algorithm operates at $30\,\mathrm{Hz}$, matching the LWIR camera's frame rate. It sustains this performance even on modest hardware (tested on a 13th Gen Intel® Core™ i7-13620H CPU). The full set of parameters used for our method is reported in this repository.[1]

We evaluated the YOLOv8-based methods using the same 2-track datasets. This process involved two configurations: first training on T1 and testing on T2 (T1→T2), then reversing the roles (T2→T1).

To create an independent test set for each configuration (e.g., T1→T2), we partitioned the source training data (T1). Specifically, we allocated 80% of T1 for training (T1a) and used the contiguous 20% as a temporal validation set (T1b). This temporal split prevents data contamination between the training and validation subsets. T1b served as the validation set for monitoring performance and applying early stopping. The entire second dataset (T2) was reserved for the final performance evaluation. For the case of T2→T1, we used the same splitting strategy but with reversed datasets (e.g. Training and test on T2a, T2b and test on T1).

We trained the model on an NVIDIA RTX 6000 GPU for a maximum of 50 epochs with a batch size of 32.

For YOLOv8 P1, we used the Stochastic Gradient Descent (SGD) optimizer with a learning rate of 0.01 and an early stopping patience of 20 epochs on the validation loss. For YOLOv8 P2, we used the AdamW optimizer with a learning rate of 0.0003 and an early stopping patience of 20 epochs on the validation loss. All other hyperparameters were kept at their default values from the YOLOv8 framework.

After training our models, for each model, we select a confidence threshold that maximizes the F1 score on the validation set. This threshold is then fixed and applied during inference on the test set to classify detections as either positive or negative for performance evaluation.

4. RESULTS

This section compares our proposed pipeline with the YOLOv8 baseline. All metrics are averaged over ten independent runs to ensure statistical robustness.

4.1 Quantitative Analysis

Table 2 summarizes the cross-track detection results. The discussion below focuses on the relative behaviour of the two methods across the test sets; raw values can be found in the table. As shown in Table 2, when training on T1 and testing on T2 (T1→T2), the proposed algorithm improves on median recall and on precision, thus achieving the highest F1 score among the four configurations. On the other hand, in (T2→T1), YOLOv8 P2 attains slightly higher precision and a comparable F1 score.

The trend reflects the underlying data: T1 offers clearer thermal signatures and less clutter, providing stronger supervision than the more challenging T2. This may be partially due to its earlier recording time, enhancing thermal contrast, and greater vegetation coverage in

[1] https://github.com/mariomlz99/pfm1-thermal-detection

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Table 2. Average detection results over 10 runs for all methods (cross-track setting). Metrics: average true positives (aTP), false positives (aFP), false negatives (aFN), Precision (aPrecision), Recall (aRecall), and F1 score (aF1). YOLOv8 results also include average confidence threshold (aConfidence; see Section 3.4).

Train→Test	Method	aTP	aFP	aFN	aPrecision	aRecall	aF1	aConfidence
T1→T2	Ours F1	1100.9	**721.9**	2293.2	0.604	0.324	0.422	—
	Ours F2	**1263.7**	781.9	**2132.3**	**0.618**	**0.372**	**0.464**	—
	YOLOv8 P1	947.3	1026.3	2449.7	0.530	0.279	0.339	0.209
	YOLOv8 P2	973.7	1755.8	2423.3	0.429	0.287	0.315	0.155
T2→T1	Ours F1	1183.3	5565.8	3329.7	0.176	0.262	**0.210**	—
	Ours F2	**1246.0**	8404.0	**3267.0**	0.131	**0.276**	0.177	—
	YOLOv8 P1	995.6	5486.2	3517.4	0.187	0.221	0.192	0.180
	YOLOv8 P2	861.1	**3263.3**	3651.9	**0.275**	0.191	0.208	0.245

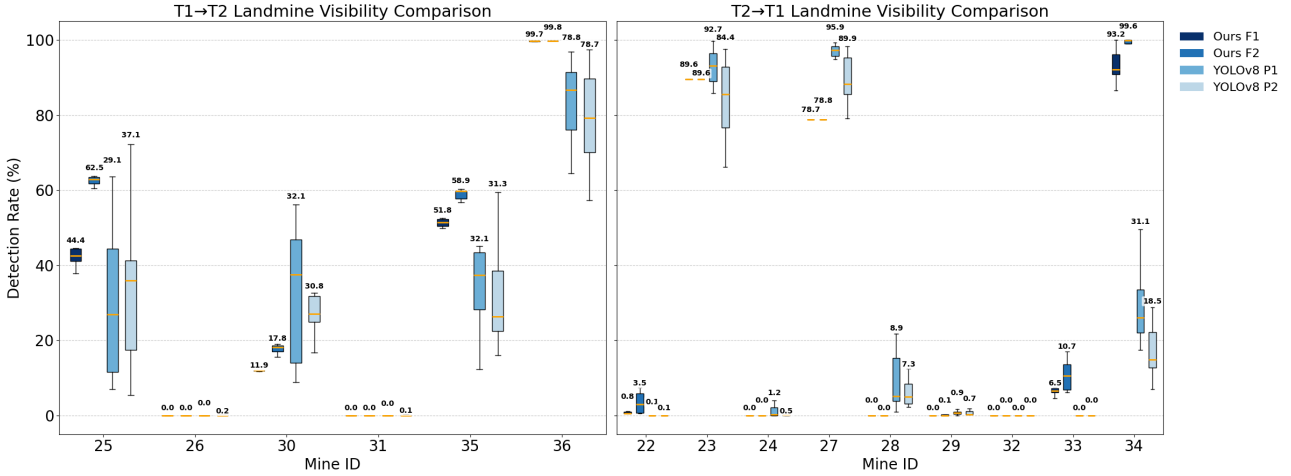


Fig. 2. Detection rates per PFM-1 landmine in cross-track evaluations (boxplots). Left: training on Track 1, testing on Track 2 (T1→T2). Right: inverse (T2→T1). Each mine ID includes results for Ours F1/F2 and YOLOv8 P1/P2. Orange lines show medians; boxes/whiskers indicate distribution.

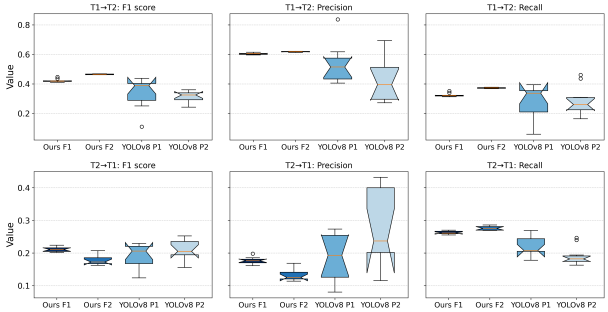


Fig. 3. Comparison of detection performance across domains using boxplots for three metrics: F1 score, Precision, and Recall. Top: training on Track 1, testing on Track 2 (T1→T2); bottom: inverse (T2→T1). Each group shows Ours F1/F2 and YOLOv8 P1/P2. Orange lines mark medians; notches show 95% confidence intervals.

T2, which may have occluded some signatures.

4.2 Qualitative Analysis

Analyzing the videos frame by frame inflates both false-positive (FP) and false-negative (FN) indicators: every missed detection in a frame is logged as an FN and every misleading trigger is logged as an FP. Consequently, detecting a mine as a true positive (TP) in only n % of the frames in which it is visible produces $(100 - n)$ % FNs. Moreover, the proposed gradient-based pipeline responds to subtle thermal fluctuations and background motion. In thermally polluted frames, where heat accumulates along structural edges or bright background regions, the pipeline may over-trigger, increasing FP counts. While this limits precision, it simultaneously supports more aggressive recall, which is often preferred in safety-critical applications like landmine detection.

4.3 Visibility-Based Analysis

Figure 2 breaks the results down by mine ID and therefore links detection performance to thermal visibility.

The two tracks differ mainly in how many mines exhibit a usable thermal signature. T2 contains six PFM-1 landmines; two never become thermally visible. YOLOv8 occasionally "detects" one of these invisible targets with very low confidence (0.1–0.2 %), probably by matching background clutter that the association logic labels as a TP. T1 hosts nine landmines; only ID 32 is completely invisible in the thermal domain, and ID 29 is marginally visible and sporadically detected. Hence, training on T1 and testing on T2 (T1→T2) benefits from a richer and more representative training set, which improves generalization even to partially occluded mines. The reverse split (T2→T1) suffers because two of the six training targets provide no thermal information at all.

Both detectors succeed on fully visible mines (e.g., IDs 23, 27, 36) and struggle once thermal contrast drops below the noise floor. Performance, therefore, declines sharply for heavily occluded or thermally invisible mines,

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independent of the underlying model.

4.4 Metric Distributions Across Models

Figure 3 summarizes the per-run distributions of Recall, Precision, and F1 score with notched boxplots; each notch spans the 95 % confidence interval (CI) of the median, so non-overlapping notches indicate significance at $\alpha = 0.05$.

Recall—Ours F2 attains the highest median recall, most notably in the T1→T2 split, suggesting better performance under occlusion. Both YOLOv8 variants, however, struggle to keep pace on both tracks. Optimizing the F2 score emphazies recall by imposing a greater penalty on FNs. Consequently, the reduction in FN occurences yields fewer missed detections, a critical requirement in humanitarian demining applications [5].

Precision—YOLOv8 achieves superior precision, reflecting fewer FPs, particularly in the T1 domain, which is characterized by a higher density of visible mines.

F1 score—The proposed pipeline retains the top median F1 score in T1→T2; the margin tightens in T2→T1 as its FP rate increases. The non-overlapping CIs for recall and F1 in T1→T2 underscore the advantage of Ours F2 when annotated data are limited.

5. CONCLUSION

We presented a feature-based pipeline for detecting PFM-1 AP mines in LWIR imagery. Due to the scarcity of high-quality thermal data, deep networks often struggle to generalize; our method instead leverages classical computer vision. Hyperparameter tuning with F1 and F2-weighted objectives prioritizes high recall, essential in humanitarian demining. Trained on clearer T1 data and tested on cluttered T2 (T1→T2), our pipeline surpasses YOLOv8 in recall and F1/F2 scores, better generalizing to partially occluded mines. In the reverse split (T2→T1), YOLOv8 achieves higher overall F1, though our system records fewer false negatives. These results suggest that a lightweight, interpretable pipeline can complement large neural detectors, especially when thermal data are limited or target visibility varies.

Future work includes synthetic LWIR augmentation and unsupervised domain adaptation across vegetation types, times of day, and datasets with temperature-driven variation.

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